

Samuel Tribble

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EDUCATION

Harvey Mudd College

Claremont, CA

B. S. Computer Science

Class of 2027

- Coursework: CSCI035: Computer Science for Insight (2024), PHIL179E: Philosophy of Artificial Intelligence (2024), CSCI070: Data Structures / Program Development (2025), ENGR191: Advanced Problems in Engineering (2024-2025).
- Awards: Cobell Undergraduate Scholarship recipient for the 2024-2025 and 2025-2026 academic years.

COMPUTER SKILLS

Python, C++, MATLAB, Java, SQL, Ubuntu (20.04), ROS 2 (foxy/humble), Gazebo, RViz2, VS Code, Git/GitHub, hardware design with Raspberry Pi microcontrollers and Radio-frequency (RF) sensors, Blender, Unreal Engine, Jupyter Notebook, Excel.

COMPUTER WORK

Pilgrim Rover LiDAR Noise Injection (ROS 2 + Gazebo)

- Designed communication between nodes where a ROS 2 Python node subscribes to a ROS 2 topic node /scan from a TurtleBot3 burger, injects Gaussian noise, and publishes the output to a ROS 2 topic node /scan_noisy.
- Wrote data analysis software with Python, NumPy, Pandas, and Matplotlib to generate a histogram of rover LiDAR data sampled in normal weather conditions vs hazardous weather conditions.

ECHO - A 3D Autonomous Rover Simulator (ROS 2 + Unreal Engine)

- Leveraged existing high-fidelity 3D models of Martian terrain and a rover that accepts manual input to design a prototype test bed.
- Extended the project base with C++ and Python to integrate ROS 2 and vision tools (e.g., YOLO).

PROFESSIONAL EXPERIENCE

Harvey Mudd College

Claremont, CA

Undergraduate Research – Research Fellow and Mentor

Aug. 2024 – present

- Developing a digital test bed with the Unreal Engine and ROS 2 to simulate autonomous Lunar/Martian rover tasks and evaluate vision models (e.g., YOLO for object detection on a 3D model of a radio game asset).
- Researching radio-frequency (RF) Ultra-wide Band (UWB) sensors and engineering hardware simulations of UWB networks in degraded Lunar/Martian environments (e.g., rover localization with a JetBot-mini or TurtleBot3 driving along a planned route in a sandstorm).
- Co-authored an academic paper on the development of large-language models (LLMs) for generating synthetic datasets of Lunar/Martian terrain classes.
- Mentored students in undergraduate research and developing deliverables in an engineering context.
- Co-authored localization algorithms with Python, NumPy, Pandas, Matplotlib and scikit-learn machine learning tools that bound/quantify the uncertainty in output from Ultra-wide Band (UWB) sensors.

INTERESTS

Aerospace, robotics, game modding, Anishinaabe philosophy, skateboarding, rock music, chocolatiering, DUCK! (a captain of the HMC improv club), investing/trading on the U.S. stock market.